

Received 27 September 2025, accepted 12 October 2025, date of publication 20 October 2025, date of current version 30 October 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3623944



AI-Driven Software Test Automation: An AI4SE-Oriented Survey of Techniques, Tools, and Challenges

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This work was supported in part by the Foundation for Science and Technology (FCT), Portugal, through the project “Intelligent System for Mapping Cultural Heritage Markers in Play Behaviour,” under Grant PRT/BD/155023/2023; and in part by FCT under Project UIDB/50008/2020.

• **ABSTRACT** The growing complexity of software systems and the need for more rapid, high-quality software releases have created the need for intelligent and automated testing mechanisms. Drawing on AI-driven test automation of software in the larger context of Artificial Intelligence for Software Engineering, this survey discusses how artificial intelligence methods are used to augment and automate multiple stages of the software test cycle, ranging from test planning, generation, execution, repair, and maintenance. We offer a systematic overview of 76 industrial tools and peer-reviewed research and examine the contribution of Machine Learning, Deep Learning, Large Language Models, Reinforcement Learning, and other paradigms of artificial intelligence to modern test automation pipelines. The paper suggests a lifecycle-oriented taxonomy, aligning industry with research with industrial practices, and offers structured visual analyses to map current trends, limitations, and emerging research opportunities. Special focus is awarded to untapped